TRADITIONAL MARKETS · OPERATIONS GUIDE · AUGUST 2026

Market Bot
Paper Desk Manual

A complete picture of the bot as it is today: the paper account, the three sleeves (Slow, Snap, Pulse), how prices are fetched, and how to run the desk. Written for an IT reader, not a trader.

Field	Value
Product	Market Bot / Paper Desk
Venue (paper)	Public prices only. No broker. No exchange account.
Mode	Paper only. Virtual $1,000. No API key, no wallet, no live order.
Slow	Monthly dual momentum: SPY vs EZU, or cash. ~80% of the book. No stop.
Snap	QQQ 2-day washout (RSI). ~8% of the book. ATR stop. 1–5 sessions.
Pulse	BTC / ETH 4-hour breakout. ~12% of the book. ATR stop + trail.
UI	http://localhost:3001
API	http://localhost:8002
Document date	13 August 2026

Read this first. The bot never sends a real order. Every fill is a simulation against a public last price plus a small fee and slippage. A green equity curve on paper is not a reason to open a broker. This is not financial advice. This project is separate from the Polymarket weather desk.

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1. What this thing is

This is a small program that runs on your machine in Docker. It watches public prices for two stock funds and two cryptocurrencies, applies three fixed rulebooks, and **pretends** to trade. Nothing leaves your laptop. There is no bank login and no broker.

Think of it as a flight simulator. The weather outside is real (live ETF and crypto prices). The plane is fake (a spreadsheet that says you have $1,000).

It keeps one ledger and three sleeves:

- **Slow.** Most of the money. US vs Europe stock baskets, once a month.
- **Snap.** A small, faster stock book on Nasdaq-100 (QQQ). Buys a crash-that-looks-like-a-dip, sells in a few days.
- **Pulse.** A small crypto book. Bitcoin and Ether, 4-hour breakouts, hard stop.

A background **worker** wakes about every 30 seconds. A website at `localhost:3001` is the desk. Postgres stores the fake ledger. Restart Docker and the ledger is still there (unless you use `down -v`).

Ports 3001 and 8002 were chosen so this desk can sit next to the Polymarket paper bot (3000 / 8001).

2. Words you will see

Word	In plain language
ETF	A listed fund that holds a basket. SPY = 500 US companies. EZU = euro-area companies. QQQ = Nasdaq-100 (big tech-heavy US names).
Sleeve	A pocket of the same $1,000 with its own rules. Slow, Snap, Pulse.
Last / mark	The latest public price we have. Used to value open positions.
Fill	A simulated trade. “We bought 1.1 SPY at 772.”
Stop	A price under the position. If last hits it, Snap or Pulse sells. Slow has no stop.
Trail	A stop that only moves up as the trade goes your way (Pulse).
ATR	Average True Range: how much this thing usually swings. Wider swing = wider stop.
RSI(2)	A 0–100 “was this just smashed?” gauge on the last two days. Very low = washout.
SMA / average	Simple moving average. 200-day = long-term trend. 10-month = Slow’s “is this still up?” line.
12–1	Return from a year ago to a month ago. Last month ignored because it is noisy.
Latent	Unrealized: (mark – buy price) × quantity. Not yet locked in. Ignores fees.
Realized	Locked-in P&L after a sell, after buy and sell fees.
Today P&L	Equity now minus equity at the start of the UTC day.
Kill switch	Stops the worker from placing new paper trades. Does not flatten positions.
Daily halt	If the day is down more than 5%, pause new trades until the next UTC day.

3. The paper account

One virtual account, id = 1, starting at $1,000 cash. Equity = cash + open positions marked at last price. Today P&L and latent are **not** the same number: today includes fees already paid and the move since midnight UTC; latent is only mark minus average on open lots.

Reset on the Control screen wipes fills, decisions, positions, and equity points, and puts $1,000 back. The `pgdata` Docker volume survives `compose restart` and `down`. `down -v` erases it.

4. Slow strategy (monthly ETFs)

Job: stay in the stronger of two stock markets if that market is still in a long uptrend. Otherwise sit in cash. About **80%** of the book. No stop-loss. One decision per month (or on first start).

Each month the bot asks two questions:

- Is this ETF's last **closed** US session above its ~10-month average? If no, it is not eligible.
- Of the eligible ones, which had the better return from 12 months ago to 1 month ago (12–1)? Hold that. If none, cash.

Default names: **SPY** (S&P 500) and **EZU** (euro area). It fills at the current public quote plus 2 bps slip and 5 bps fee. The *decision* uses yesterday's close so today's unfinished session cannot leak into the signal.

Judge Slow over months. A –$4 day is noise. A 5% stock stop would shake this book out of the very trend it is trying to ride.

5. Snap strategy (aggressive QQQ)

Job: buy a short, violent dip in Nasdaq-100 **only if the long trend is still up**, then get out in a few days. About **8%** of the book. This is the aggressive traditional sleeve — smaller and faster than Slow, stocks not crypto.

Rules, in order:

- Trade only **QQQ** (Nasdaq-100 ETF).
- Enter on a **closed** daily bar when RSI(2) is at or below 10 (a 2-day washout) **and** price is still above the 200-day average (do not catch a falling bear market).
- Initial stop = entry – 2.5 × ATR(20). Checked against the live quote every 30 seconds.
- Exit if price closes back above the 5-day average (the bounce happened), or after 5 sessions, or if the stop is hit.
- Size so the stop is about 0.75% of total equity, then cap the whole sleeve at 8%.

Holds are usually 1–5 trading days. Many trades will be small losers (the stop). That is the design. Do not retune Slow because Snap had a good week.

6. Pulse strategy (crypto)

Job: ride a Bitcoin or Ether push if a 4-hour bar closes at a new high, and cut if it fails. About **12%** of the book. Long only. No borrow.

- Enter when a **finished** 4-hour bar closes above the prior 20-bar high.
- Stop = entry – 2.5 × ATR(14). Trail = close – 3 × ATR (only moves up).
- Also exit if a 4-hour bar closes under the prior 10-bar low.
- Live stop is checked every 30 seconds. New entries / trails only on a new closed 4-hour bar.
- After a stop, stay flat until a 4-hour close is back **inside** the channel (no re-entering the same failed breakout).
- Do not enter if the live price is already through the stop.

- Risk about 1% of total equity to the stop, then cap the sleeve at 12%.

A “good” Pulse book loses often and wins fat. Long quiet periods are normal. It cannot empty Slow.

7. Data sources and how often they are called

What	Source	How often
Worker loop	Local	Every 30 seconds (Control: Loop)
SPY / EZU / QQQ history	Stooq CSV, Yahoo 2y fallback	First boot, then at most every 3–12 hours
SPY / EZU / QQQ last	Yahoo 5-day chart	About once an hour
BTC / ETH last	Binance ticker	Every 30 seconds
BTC / ETH 4h candles	Binance klines	Only if we lack the last closed 4h bar

No API keys. Binance will not run out at this rate. Yahoo is unofficial and is the only source that might briefly return 429 if hammered; the worker then keeps the last mark. The UI never calls these sites — only the worker does.

8. The desk (screens)

Tab	What it is
Overview	Equity, today, latent, realized, hit rate. Chart with Today / 7d / 30d / All. Three rails. Recent decisions and fills.
Slow	SPY vs EZU scores and the monthly holding.
Snap	QQQ RSI, stop, washout / bounce tape.
Pulse	BTC / ETH channel, ATR, armed / disarmed, stop.
Trades	Open positions first, then closed. Fills and decisions.
Analysis	Last 20 completed trades. Notes only fire when the sample is large enough.
Control	Start / pause / kill / reset. Sleeve sizes and stops.

9. How to run it

```
cd "/Users/damien/Documents/02 - Dev/market_bot"
cp .env.example .env
docker compose up --build -d
# UI  http://localhost:3001
# API http://localhost:8002/health
```

Compose project name is `market_bot` so it does not collide with the Polymarket stack. Kill switch and daily halt are paper-only. Reset bankroll from Control.

10. Settings

Defaults live in `.env` and in the Control page. New keys are inserted if missing; existing values are not overwritten on restart. After a Reset, Slow may buy again on the next first-run month clock.

Knob	Default	Meaning
slow_sleeve_fraction	0.80	Share of equity for monthly ETFs
snap_sleeve_fraction	0.08	Share of equity for QQQ
pulse_sleeve_fraction	0.12	Share of equity for crypto
snap_rsi_buy	10	Enter Snap when RSI(2) is at or below this
snap_max_days	5	Force Snap exit after this many sessions
pulse_stop_atr / trail	2.5 / 3.0	Pulse stop and trail width
daily_loss_halt	0.05	Pause new trades if the UTC day is down 5%
poll_interval_seconds	30	Worker wake-up. Does not change monthly / 4h clocks

11. When something looks wrong

You see	Usually means
SPY open, no stop	Slow is working. There is no stock stop on purpose.
Pulse flat for days	Price is inside the 20-bar high. Correct.
Snap flat for weeks	No 2-day washout, or QQQ is under the 200-day average.
Today ≠ latent	Fees, UTC midnight, or a closed trade. Same as the other desk.
Yahoo 429 in logs	Quote skipped; last mark is kept. History is not pulled every 30s.
API offline on 3001	API is 8002. Confirm compose project market_bot is up.

12. What it will not do

- Place a live broker order. There is no Interactive Brokers (or any) adapter.
- Trade single stocks, futures, or raw materials. Paper ETFs and two coins only.
- Short Pulse or Snap. Long only.
- Replace a lawyer, a tax advisor, or a licensed broker.
- Make you eligible for any venue. That is a person-and-country question, not a server question.

Companion document: `procedure_live.pdf` — what going to real money would actually require.